

ISABELLA CHANDLER

Biomedical Engineering | Medical Devices | Quality Engineering | Healthcare AI
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PROFILE

Biomedical Engineering senior with hands-on experience in biomedical instrumentation, medical-device prototyping, quality-focused device operations, healthcare AI concepts, and technical project leadership. Quality Science Education Certificate holder interested in medical devices, imaging, quality/test engineering, and product development.

EDUCATION & QUALITY TRAINING

B.S. Biomedical Engineering | Lawrence Technological University, Southfield, MI *Expected May 2027*
GPA: 3.16/4.00 **Current coursework:** Biomechanics + Lab, Tissue Engineering + Lab, Foundations of Medical Imaging, Introduction to Nanotechnology, BME Projects I
Recent coursework: Bioinstrumentation + Lab, Biomaterials, Medical Device Design, Robotics & Society, Introduction to BME Projects
Quality Science Education Certificate | Pathway for Patient Health *Aug 2025*
Training: FDA quality principles, GMP, validation, risk assessment, change control, data integrity, and quality systems for pharmaceutical and medical-device environments.

EXPERIENCE & LEADERSHIP

Help Desk Assistant | Lawrence Technological University *Jan 2025-Present*
• Completed **2,100+ logged laptop inspections/cleanings and fault checks** for university redistribution; verify readiness across Lenovo T1/T2/T3, x360, and Fujitsu systems and document issues found during imaging or QA.
• Run imaging workflows for Fujitsu and Lenovo systems, execute deployment scripts, stage devices/chargers for dual-enrollment and semester distributions, and maintain HP/Ricoh printers through paper checks and toner replacement.
• Trusted to independently manage back-room operations during extended summer shifts while balancing inventory, imaging, cleaning, charging, and deployment goals; worked approximately 20 hrs/week during Spring 2026 alongside full-time coursework.
Director of Programming, Executive Board | LTU Student Government *Aug 2025-May 2026*
• Directed planning and execution of a **\$10,000+ End-of-Year Celebration**, coordinating vendors, food trucks, inflatables, facilities, schedules, purchasing, and day-of logistics; previously served as Senator and committee leader.
Biomedical Engineering Lab Assistant | Lawrence Technological University *Apr-Aug 2024*
• Prepared BME teaching spaces and equipment for summer-camp activities; organized supplies and Excel inventory and completed routine lab-readiness/safety tasks.

SELECTED ENGINEERING PROJECTS

Mopec Tissue Insight | Company-Sponsored Senior Design *2026-2027*
• Contribute to early-stage development of an AI-assisted gross-tissue imaging workflow through dataset review, image-quality/preprocessing planning, annotation strategy, train/validation/test structure, and model evaluation planning.
HeadUp Posture-Feedback Wearable | Bioinstrumentation
• Designed and tested an accelerometer-based posture wearable using calibration and threshold logic with CircuitPython/embedded feedback; evaluated sustained-slouch detection, activation delay, comfort, and repeatability.
Pressure-Sensing Mobility Aid | Medical Device Design PPA Draft
• Co-developed a smart replacement tip for crutches, canes, and walkers using load-cell sensing, Bluetooth data transfer, threshold-based LED feedback, and a patient/clinician data interface; co-authored the PPA-style design package with system drawings, technical description, and claims.
Medical Device Design Projects | BME 4113
• Built/tested an ECG acquisition chain using an AD620, op-amps, passive filters, and Arduino; completed regulatory/design work including a 510(k)-style surgical-mask summary, device drawings, and sensor-focused labs.
ConnectUs Accessible Dice Tower | Entrepreneurial Engineering Design Studio
• Translated customer needs into requirements and iterated an accessibility-focused prototype using concept selection, Arduino/servo control, rapid fabrication, soldering/wiring, testing, and user feedback.

TECHNICAL SKILLS

Engineering & analysis: MATLAB, COMSOL, CircuitLab, Excel, AutoCAD/CAD, mechanical testing, analog filters, prototyping, 3D printing, soldering/wiring, sensor calibration/testing
Programming & embedded: Python/CircuitPython, C#, Arduino, BLE, CRICKIT, microcontrollers, accelerometers, servo control
Quality & regulatory: quality inspection, validation fundamentals, risk assessment, GMP/FDA quality principles, technical documentation, data integrity, 510(k) familiarity, PPA/patent-style documentation
AI & tools: Gemini AI Studio, GitHub, Microsoft 365; Spanish (Advanced)